sensonomic

# D10: Assessment of annual fruit yields

# Shea tree productivity

This document first presents existing secondary data from five studies on average shea tree productivity monitored over multiple years in several West and East African countries. In a second part, current knowledge about factors influencing shea tree productivity is reviewed.

## 1. Results of shea tree productivity studies

Summary tables 1 to 5 show the wide variation in average productivity of shea trees from year to year in any location, which is expected from a semi-domesticated tree species managed in the wild. While not presented here, individual tree yields for any one year also vary widely from tree to tree (Table 3 over 5 years), commonly by a factor of 5 or more between the population average and the highest producer in a population. A good year in one place does not mean that production will be comparatively high in another location in the same year (Table 2). Land management practices also affect shea productivity and shea trees are most productive in cultivated fields rather than fallows or forests, yet patterns of inter-annual variability can also differ by land-use type (Table 4).

There is considerable tree to tree diversity in morphological characteristics, with various authors reporting the following intervals:

- fruit length: 4-5 cm, occasionally 8 cm (Hall et al., 1996); 3.3-4.7 cm (Kelly and Senou, 2017)
- fruit diameter: 2.5-5 cm (Hall et al., 1996); 2.9-4.2 cm (Kelly and Senou, 2017)
- fruit weight: min 10g, mean 20-30g, max 57g (authors in Hall et al, 1996); 15-60g (Maranz et al, 2004); 18.7-53.7g (Kelly and Senou, 2017)
- pulp weight: 8.6-41.2g (Kelly and Senou, 2017)
- fruit pulp vs seed ratio: min 33%, mean 55%, max 75% (Ruyssen, 1957)
- nut length: 2.8-5 cm
- nut diameter: 2.2-3.5 cm
- nut fresh weight: min 4.7g, mean 8-10g, max 16.0g (Hall et al., 1996); 9.6-12.6g (Kelly and Senou, 2017)
- nut dry weight: 3.8-6.0g (Maranz et al., 2004)
- kernel weight: min 2.6g, mean 4.4-5.7g at 7% moisture content, max 6.6g (Ruyssen, 1957) or dry kernels 3.9-6.2g in Benin (Schreckenberg, 1996)
- husk weight

- kernel fat content: 20-50% (Maranz et al., 2004)

Reported conversion ratios include the following:

- fresh kernel: 15-20% of fresh fruit weight (Hall et al, 1996)
- kernel dry weight: 29-47% of fresh nut weight (Vuillet, 1911)
- nut dry weight: 25% of fresh fruit weight (Maranz et al, 2004)

The Ex-Ante Carbon Balance Value Chain tool developed in the GSA-FAO study uses dry kernel means per tree of 3 and 4 kg. Lovett (2004) used an estimate of 5kg of dry kernel per tree.

**Table 1. Individual tree production of shea fruits in the Zoundweogo Province of Burkina Faso, 1993-1995**

Burkina Faso

| | **Trees** | **Fruits** | **Fruits** | **Fruits** | **Avg fruits** | **Avg fresh nut kg/tree** |
|---|---|---|---|---|---|---|
| | | **1993** | **1994** | **1995** | | |
| Gomboussougou | 53 | 223 | 1,019 | 1,047 | 763 | 5.7 |

**Table 2. Individual tree production of shea fruits in three sites on a N-S gradient in Burkina Faso, 2008-2010 (Bastide, unpublished data)**

Burkina Faso

| Locations | **Trees** | **Fruits** | **Fruits** | **Fruits** | **Avg fruits/tree** |
|---|---|---|---|---|---|
| | | **2008** | **2009** | **2010** | |
| Guibaré | 60 | 237 | 329 | 692 | 419 |
| Sobaka | 60 | 7,182 | 694 | 3,497 | 3,791 |
| Niangoloko | 60 | 768 | 1,822 | 1,971 | 1,521 |

**Table 3. Individual tree production of shea fruits in Tengrela, northern Côte d'Ivoire, 1998-2002 (Diarrassouba, unpublished data)**

Northern Côte d'Ivoire

| | **Trees** | **Fruits** | **Fruits** | **Fruits** | **Fruits** | **Fruits** | **Avg. fruits/tree** |
|---|---|---|---|---|---|---|---|
| | | **1998** | **1999** | **2000** | **2001** | **2002** | |
| Tengrela | 124 | 1,598 | 1,384 | 770 | 1,074 | 2,024 | 1,370 |

**Table 4. Individual tree production of shea fruits in the Tanguiéta and Materi communes, Northern Benin, 2014-2015 (Aleza, 2015)**

Benin

| Land use type | Fresh fuit mass (kg) | Stdev | Number of fruits | | | |
|---|---|---|---|---|---|---|
| | | | 2014 | Stdev | 2015 | Stdev |
| Farmland | 17.4 | 11.7 | 1136 | 955 | 791 | 742 |
| Young fallow | 12.5 | 11.9 | 649 | 568 | 715 | 855 |
| Old fallow | 11.4 | 11.0 | 620 | 519 | 916 | 1018 |

**Table 5. Individual tree production of shea fruits in four districts of northern Uganda, 2010-2011 (Byakagaba, unpublished data)**

Uganda

| | Trees | Fruits | Fruits | Avg. fruits/tree |
|---|---|---|---|---|
| | | 2010 | 2011 | |
| Katakwi | 35 | 298 | 1,160 | 729 |
| Lira | 35 | 165 | 479 | 322 |
| Moyo | 38 | 203 | 703 | 453 |
| Nakasongola | 25 | 28 | 48 | 38 |

## 2. Variation and factors affecting productivity

**Inter-annual variation:** Shea production is characterized by variable cycles of good, average and bad harvests impacting export volumes and perception of sustainability of the sector. In young shea parklands of Thiougou, Southern Burkina Faso, the 3-year average nut yield of 53 trees was 2.4kg. However, average yields were almost five times higher in years 2 and 3 than in year 1, showing patterns of variable annual bearing. Alternate bearing patterns originate from carbohydrate deficit in plant organs resulting from high production during one season which generally leads to low production during the next season (Monselise and Goldschmidt, 1982). Alternating cycles can be regularly maintained over several years or be interfered with by climatic or pathological factors.

**Intra-population variation:** There was also a five-fold difference between the best producing trees and the population average, indicating the potential for selection and improvement. In the sample population, 42% of the trees were consistently very low producers and their contribution represented less than 10% of total production. High producers in two or three years represented 15% and 9% of the whole sample, but 35% and 20% of total production (Table 6). Results of a tree monitoring study over five years in northern Côte d'Ivoire show similar patterns (Table 7). Productivity thus appears to depend both on the genetic makeup of the tree as well as external factors. Tables 1a and 1b demonstrate the potential benefits of increasing of an improvement program that would increase the proportion of high-yielding trees in any shea population.

Table 6. Interannual variation of shea nut yields from *Vitellaria paradoxa* trees in Southern Burkina Faso during 1993-1995

| **3-year production** | **Number of trees Years when nut yield > annual average** | | | | **Total** |
|---|---|---|---|---|---|
| | **0/3** | **1/3** | **2/3** | **3/3** | |
| 0-3.5kg | 17 | - | - | - | 17 |
| 3.5-7.319*kg | 5 | 11 | - | - | 16 |
| 7.32-15kg | - | 7 | 3 | 2 | 12 |
| >15kg | - | - | 5 | 3 | 8 |
| No. trees | 22 | 18 | 8 | 5 | 53 |
| % trees | 42 | 34 | 15 | 9 | 100 |
| % yield | 10 | 34 | 36 | 20 | 100 |

*7.319kg is the 3-year cumulative production mean in the sample

Table 7. Interannual variation of shea nut yields from *Vitellaria paradoxa* trees in Northern Côte d'Ivoire during 1998-2002

| | **Number of trees Years when yield > annual average** | | | | | | **Total** |
|---|---|---|---|---|---|---|---|
| | **0/5** | **1/5** | **2/5** | **3/5** | **4/5** | **5/5** | |
| No. trees | 23 | 30 | 25 | 23 | 17 | 6 | 124 |
| % trees | 19 | 24 | 20 | 19 | 14 | 5 | 100 |
| % yield | 7 | 16 | 20 | 26 | 22 | 9 | 100 |

**Tree age:** Shea trees are slow-growing and long-lived. Nut production increases with tree age. It becomes significant between 20 and 30 years of age, increases until age 100 to 200 and would slowly decline afterwards. It has been estimated that shea trees may then live up to 200 or 300 years (Delolme, 1947; Ruyssen, 1957), yet no dendrochronological or carbon-dating studies have yet been reported for shea. However, no correlation was found in the Southern Burkina Faso study between fruit yield and tree diameter over the relatively limited range of 10-44 cm dbh (Boffa et al., 1996). Similarly Kelly et al. (2007) did not find any difference in flowering parameters according to tree size in their study. Yield variations linked to tree age are thus expected over the lifetime of trees but may be small over limited age intervals.

**Climatic factors:** Several authors have proposed general hypotheses relating climate to fruit productivity but these have not been investigated systematically. The contributions of climate (rainfall, temperature) and soil factors to variations in shea phenology and production need to be further quantified and disaggregated, and optimum conditions assessed.

Temperature: Higher minimum temperatures during flowering (November through February) results in higher nut yields according to Desmarest (1958). Because average minimum temperatures decrease regularly in the Sahel from November to January as the relatively cool dry season progresses, an early start of flowering would also result in higher nut production. However, in Uganda fruiting frequency (not fruiting quantity) in *ssp. nilotica* was weakly positively correlated with mean daily temperature while it was strongly negatively correlated with both relative humidity and wind speed (Okullo et al., 2004). Similarly bush fires during flowering especially later in the dry season when and the harmattan - that blows south from the Sahara across the Sahel during the cool, dry season - can adversely affect the timing and quantity of flowers produced and hence have a negative influence on production (Ruyssen, 1957). In Uganda, flowering was concentrated late in the dry season after the occurrence of most fires, so that disruption of the reproductive process and damage of reproductive organs by fire was minimized (Okullo et al., 2004).

Soil moisture: No clear pattern regarding the effect of rainfall on shea productivity has yet been identified. Delolme (1947) hypothesized that sufficient soil water reserve or the absence of drought at the time of flowering and fruit set would allow higher fruit yields. Using the example of two contrasted years, this author mentioned that an early end of the rainy season and low rainfall in the last weeks followed by a drought might be the reason behind low fruit set during the following fruiting season. However, using rainfall and nut production data of a shea population over eleven years, Ruyssen (1957) showed that there was no systematic correlation between amounts of rain fallen in one year and fruit yield the next year. The relationship between rainfall and subsoil water content that trees can draw from over the dry season, depth of water table and how this varies by soil profile type and latitude would also need to be elucidated.

Flowering ability and probability of abundant flowering over two years were higher in a north Guinean site in Mali characterized by higher rainfall, lower mean annual temperature and higher relative humidity than a compared south Sudanian parkland site (Kelly et al., 2007). The flowering period started earlier and was shorter in the more favorable climatic conditions. However, the Guinean site also appeared to have more fertile soils; yet soil conditions were not investigated. Thus, the respective impact of climatic conditions (temperature, air and soil moisture) and soil fertility factors and their interactions on shea phenology and production remain an open field of investigation for improved shea management.

Irrigation experiments of shea stands in the drier part of the range and contrasted wetter sites, for instance with water collected through locally appropriate water harvesting techniques, could help define appropriate amounts and timing of water supply for maximizing flowering, fruit set and fruit production.

**Soil type and nutrient status:** Shea is incompatible with some soil types. It avoids even temporarily flooded areas and does not occur on highly sandy or clayish soils. The species is well adapted to poor shallow soils and land suitable for rainfed crops (Ruyssen, 1957; Picasso, 1984). Soil fertility influences growth and production in that shea trees are more vigorous on deep alluvial soils than on lateritic gravelly soils and they may yield more on fertile soils in low production years. However, authors differ in their interpretation of the influence of soil fertility on the large differences in productivity that one finds in stands of mature shea trees. Delolme (1947) identified three groups of neighbouring shea trees of similar diameters and production and reported yield differences between them that were ranked similarly over two years. He attributed yield differences in these stands to soil types through an appreciation of grass height and vigour under the trees. However, the latter assessment was visual and unquantified and the undisclosed sampling metholodogy does not rule out other determining factors. Conversely, Desmarest (1958) conducted a series of auger-based soil texture assessments at various soil depths under trees of contrasted yields and found similar soil profiles. He concluded that soil type was not responsible for the large observed tree-to-tree productivity variations. The same conclusion – that soil type has little influence on shea production – was reached by Picasso (1984).

Weed control by hoeing, intercropping beans and manure application had positive effects on fruit yields in Ghana (Hall et al., 1996). N and P fertilization in nursery conducted by Dianda et al. (2009) stimulated the growth of 6-month old shea seedlings but response depended on the ratios between both nutrients. N limitations appeared to be the main constraint to biomass accumulation. When investigating the N requirements for optimal growth of shea seedlings, N dosage much lower than 12 $mg.kg^{-1}$ should be used under conditions of P shortage. External N inputs relative to P supply and possibly soil K concentration was recommended for successfully using mineral fertilizers in shea nursery production. This experiment also showed that shea is a susceptible host species to the arbuscular mycorrhizal (AM) fungus *Glomus intraradices*, yet no benefits of AM fungi inoculation to seedling growth were evident, probably because no P stress occurred.

**Pollination**: Shea is an outcrossing species (Yidana, 1994). Bees play an important role in pollination (Sallé et al., 1991) and shea pollen occurs in honey samples (Schweitzer et al., 2014). However, relatively little research has been done on this aspect.

**Girdling:** Lamien et al. (2006a) demonstrated that shea responds to girdling, a technique used to control fruit yield irregularity by causing carbohydrate and hormonal accumulation in stems and leaves located above the incision point. The technique resulted in a 100% increase in fruit production. No significant difference between girdling dates (August when fruit fall ends and late November when leaf fall starts) was observed. The technique can be recommended with the specification that incisions are made in ways to ensure rapid wound closure. Nevertheless assessment of girdling’s long term effects on trees is desirable.

**Vegetation management:** Flowering attributes in shea trees located in fields were consistently more favourable than in fallow or forest locations (Kelly et al., 2007). Thus farming activities which result in the structuring of agroforestry parklands enhance flowering of shea. Likewise average fruit yields measured in a single year were higher in agroforestry parklands (4.3kg/tree) than in neighbouring natural woodlands (1.6kg/tree) and the proportion of trees with zero fruit production was 16% in agroforestry parklands and 48% in natural woodlands (Lamien et al., 2004). Average tree diameter as well as fruit size and weight were also larger in parklands than those in uncultivated plots. These more favourable conditions probably reflects the combined positive factors of farmer selection of trees, cultivation practices on soil nutrient status, reduced tree competition, larger tree size, as well as control of grazing and bush fires.

**Parasitism:** Another potential factor affecting tree productivity is shea's vulnerability to infestation with hemiparasitic plants (Loranthaceae), including three species of mistletoe parasites *Agelanthus dodoneifolius*, *Tapinanthus globiferus* and *T. ophiodes*. Infestation rates are reported to be high in West African parklands, 95% in Burkina Faso (Boussim et al., 1993) and 81% in Nigeria (Odebiyi et al., 2004). Shea's nutritional deficit caused by the parasites is said to lead to discontinued growth, decline and death of the distal part of infested branches. As parasites can grow rapidly and insert themselves in multiple locations deflecting water supplies away from the host, infestation can cause generalized tree weakening, permanent defoliation, drying out of affected tree parts, lowered flowering and fruiting and tree death. The latter is a common sight in the Northern part of shea's range in Burkina Faso where trees are also subject to droughts and harsh winds (Boussim et al., 1993; Boussim et al., 2004).

The frequency of *Tapinanthus* infestation is considerably higher in shea trees in cultivated parklands (80%) than in protected natural woodland areas (27%) and the parasite load was higher in parkland trees (Houehanou et al., 2011). Larger trees also tend to be more heavily infested. These differences in infestation rates between parklands and natural woodlands are probably be due to the larger average size of shea trees in parkland conditions which provide greater infestation potential and higher production of parasite seed, as well as more intense dispersion by birds in the more open parkland environment. However, presence/absence of parasites did not result in differences in flowering or fruiting of shea branches (Lamien et al., 2006b) and the level of parasite infestation (1-3 vs. 10-14 parasite branches) had no significant impact of fruit production in either of parkland or natural woodland conditions (Houehanou et al., 2013). These specific results tend to alleviate concern for the potentially negative impact of Loranthaceae parasites, as a single agent, on shea fruit productivity. Yet these parasites remain a strong concern for the sustainability of northern shea tree populations which are subject to additional climatic stresses such as droughts.

**Diseases and pests:** Shea is a hardy species and its vulnerability to fungal diseases is low. Attacks by *Fusicladium butyrospermi* Griff. and Maubl. and *Pestalozzia heterospora* Griff. And Maubl. are limited to dark brown or grey spots on shea leaves (Sallé et al., 1991). However, shea also has a number of insect pests. Sallé et al. (1991) documented a diversity of herbivorous insect pests attacking various parts of the tree in West Africa. In the Southwest and Northwest of Nigeria where shea is found, Odebiyi et al. (2004) found 33 insect pest species from 17 families on shea in three ecozones. They indicated that the majority of the insects encountered was of little significance and would not warrant application of control, to the exception of *Cirina forda* (Saturnidae:lepidoptera), a major pest causing 60–90% defoliation of mature trees of *V. paradoxa* in the Southern Guinea Savannah. However, herbivory occurs after fruit production, only on old non-photosynthetic leaves and results in brand new foliation (B. Bastide, pers. comm.) and better fruit yields the following year according to farmers (P. Lovett, pers. comm.). The caterpillar provides an incidental source of food to locals. Evaluating its potential as poultry feed in Nigeria, Kenis et al. (2014) showed that it can substituted in up to 75% of the fish meal intake in broiler chick diets without negative impact on weight gain. Lamien et al. (2008) also identified a shoot and fruit borer, Salebria sp. (Lepidoptera: Pyralidae) in Western Burkina Faso which infected 49–80% of shea trees and 4-15% of fruits.

Table 2. Summary of factors influencing fruit yields of shea trees

| Factor | Effect | Reference |
|---|---|---|
| Inter-annual variation | • Average annual yields of shea stands can vary by a factor of five between years due to alternate bearing | Ruyssen 1957<br>Boffa et al. 1996 |
| Intra-population variation | • Factor of 5 or more between the best and an average producer | Ruyssen 1957<br>Boffa et al. 1996 |
| Tree age and diameter | • Yields increase over the tree's 200 year or longer lifetime,<br>• Yet minor effect over small age intervals compared to phenotypic variation<br>• No impact on flowering attributes | Ruyssen 1957; Delolme 1947 Boffa et al. 1996<br>Kelly et al. 2007 |
| Temperature | • Higher average minimum temperatures during November-February and an early start of flowering period (Oct-Dec) result in higher nut yields.<br>• Bushfires and harmattan wind/dust reduce yields. | Desmarest 1958<br>Ruyssen 1957 |
| Soil moisture (rainfall) | • No clear effect on its own. Likely interactions with soil and subsoil factors and temperature | Ruyssen 1957; Kelly et al. 2007 |
| Soil fertility | • Shea is apparently adapted to poor shallow soils and areas adapted to rainfed crops.<br>• No significant effect on productivity of matures trees on these soils.<br>• Positive effect of weed control, intercropping with legumes and manure on mature trees.<br>• N and P fertilization stimulates seedling growth in the nursery | Ruyssen 1957<br>Desmarest 1958; Picasso 1984<br>Hall et al. 1996<br>Dianda et al. 2009 |
| Plant hemiparasites | • No difference in yield or flowering parameters between different levels of infestation<br>• Strong impact on tree vigour longevity in the northern drought-affected part of the range | Houehanou et al. 2011; Lamien et al. 2006b<br>Boussim et al. 1993 |
| Pests and diseases | • Insignificant diseases and fungal attacks<br>• Insects, *Cirina forda* and a shoot and fruit borer (*Salebria sp.*) create major damage | Sallé et al. 1991<br>Odebiyi et al. 2004<br>Lamien et al. 2008 |
| Parkland management | • Flowering attributes more favourable in farmed parklands than in unmanaged forest | Kelly et al. 2007<br>Lamien et al. 2004 |

| | | |
|---|---|---|
| | • Fruit yield 3 times higher in farmland than in unmanaged forest<br>• Twice as many fruit-bearing trees in parklands as in unmanaged forest<br>• Larger average tree diameter as well as fruit size and weight parameters in parklands than in unmanaged forest | Lamien et al. 2004<br><br>Lamien et al. 2004 |
| Girdling | • Yield increases by 100%. No difference in timing of the technique (August vs. November). No knowledge of long-term impact. | Lamien et al. 2006a |